AIRTEL SYNC HACKATHON 2026 · TRACK A SUBMISSION

PROJECT UDYAM DIGITAL

Making Small Businesses Go Digital in One Sitting (< 24 Hours) — Connectivity, SaaS & Cyber Security

TEAM **CountDown_VNIT** | MEMBERS Saurabh Borikar · Anupam Ayush

₹2,450
Target CAC
-83.7% from ₹15,000

< 24 Hours
Activation Latency
-95% from 17.6 days

28.0%
Funnel Conversion
3.8x uplift from 7.36%

0 Calls
Status Chases
100% self-serve tracking

OFFICIAL SUBMISSION DELIVERABLES & INTERACTIVE LINKS (CLICKABLE)

Product Demonstration Video

https://drive.google.com/file/d/1TwcPl2c8KbH7sJIPwkrbOD9QJjv4vQ7D/view?usp=drive_link

Voice & Smart KYC Demo · Full end-to-end walkthrough of all 5 modules & DPDP compliance

GitHub Source Code Repository

https://github.com/saurabhborikar13/Airtel-sync

Production Flask app, trained RF propensity model & test suite

Market Opportunity & The SMB Friction Problem

The MSME Market Opportunity

63M+ Micro, Small & Medium Enterprises

Backbone of Indian GDP (30%) and employment (110M+ workers). Only ~15% currently possess managed enterprise digital infrastructure.

₹2.5 Lakh Crore Connectivity & SaaS TAM

Rapidly shifting from unorganized consumer broadband to bundled B2B connectivity, cyber safety, cloud storage, and POS systems.

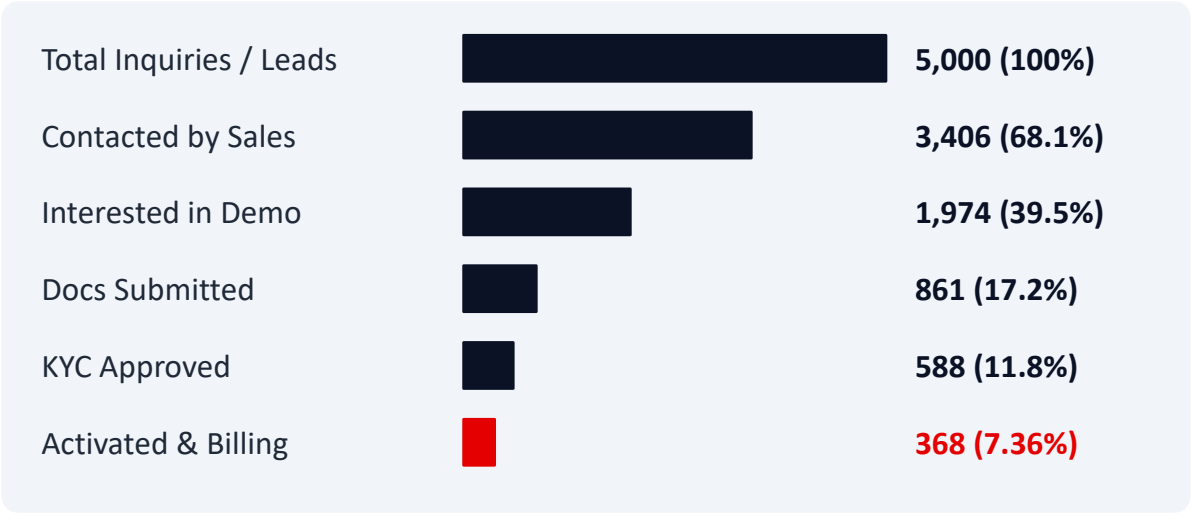
Airtel's Strategic Advantage

Unmatched nationwide fibre footprint, 5G standalone edge, trusted enterprise relationship, and comprehensive business product ecosystem (25+ B2B products).

The Growth Bottleneck

Massive top-funnel demand is throttled by complex paper onboarding, multi-visit document chases, and non-transparent pricing.

Current Funnel Reality (smb_leads.csv Baseline)



17.6 Days Latency

Physical form submissions & manual back-office verifications

₹15,000 High CAC

Multiple field RM visits, paper courier, high churn leakage

3.0 Status Chases

Frustrated SMB owners calling reps repeatedly for updates

11.2% Early Churn

Missed SLA expectations, fragmented multi-vendor setups

Target Segments & 4 Pre-Configured Value Bundles

Micro Retail

55% of Leads

₹1,998 / mo

10% Bundle Discount

Target Persona: Kiranas, apparel shops, local pharmacies, food joints

Included Solutions:

- Airtel Biz Fibre 100 Mbps (Unlimited)
- Smart Cloud POS & Soundbox
- Secure Airtel Guest Wi-Fi Portal
- Digital Payment Settlement Integration

Business Impact: Fast checkout, customer Wi-Fi marketing, zero downtime for UPI payments.

Professional Offices

25% of Leads

₹2,698 / mo

12% Bundle Discount

Target Persona: Chartered Accountants, law firms, doctors, consultants

Included Solutions:

- Airtel Biz Fibre 300 Mbps High Speed
- Static IP Address (for ERP & Tally)
- Airtel Cloud Backup (1 TB Encrypted)
- DNS Cyber Threat Protection Shield

Business Impact: Remote client data access, compliant automated daily backups, secure audits.

Small Multi-Site

15% of Leads

₹3,798 / mo

15% Bundle Discount

Target Persona: Retail chains, diagnostic labs, boutique gym branches

Included Solutions:

- Airtel Biz Fibre 300 Mbps Primary
- Automated 4G/5G Wireless Failover SIM
- Airtel Cloud CCTV (8-Channel, 30d retention)
- Centralized SD-WAN Hub Management

Business Impact: 100% operational uptime, centralized surveillance, unified multisite billing.

Small Manufacturers

5% of Leads

₹6,900 / mo

18% Bundle Discount

Target Persona: Auto-component workshops, fabrication, food packaging

Included Solutions:

- 1:1 Dedicated Internet Leased Line (10 Mbps)
- Industrial IoT Machine Telemetry Gateway
- Next-Gen Hardware Firewall (NGFW)
- Priority 24/7 Enterprise Mission SLA

Business Impact: Guaranteed low latency, machine telemetry logging, perimeter cyber security.

Solution Architecture & One-Sitting User Journey

4-Tier Scalable Solution Architecture

1 Presentation Layer

Responsive HTML5/JS Web App, Airtel Theme, Web Speech API (Multilingual Voice in EN/HI/TA)

2 Application & API Gateway

Modular Flask REST Framework, Advisor Engine, KYC Verification Service, Order Lifecycle Tracker

3 Intelligence & Decision Layer

GenAI Rule Engine, Random Forest Propensity Classifier (AUC 0.645, 12 features), 12-Check KYC Engine

4 Data & Compliance Vault

Mock GSTIN & Udyam Registries (6,200+ records), DPDP Act 2023 Consent Audit Logs, AES-256 Storage

One-Sitting Customer Journey (< 10 Minutes)

1 Step 1: Voice / Text Needs Discovery

SMB owner speaks in Hindi, Tamil, or English; AI matches archetype & formulates customized proposal in <7 seconds.

2 Step 2: Instant Transparent Quote

Auto-applies bundle discounts (10–18%), computes 18% GST itemization, and outputs branded PDF summary & WhatsApp quote.

3 Step 3: Instant Registry Pre-fill

Live lookup against GSTIN & Udyam registries pre-fills enterprise name, constitution, and authorized signatories without manual entry.

4 Step 4: Smart Specimen KYC Verification

Rule-based document validation checks blur, skew, 90-day recency, Mod-36 checksum, and signatory authority in seconds.

5 Step 5: Order Dispatch & Real-Time Tracking

Generates Order ID, dispatches hardware, allocates field installation engineer, and provides customer self-serve tracker.

Core Innovations: GenAI Advisor & Smart KYC Engine

GenAI Multilingual Advisor

Voice-First Consultation (Web Speech API)

Micro-entrepreneurs often prefer speaking over typing. Supports real-time speech-to-text in English, Hindi, and Tamil.

Archetype-Driven Semantic Engine

Maps informal prompts (e.g., 'mujhe payment machine aur fast internet chahiye') into structured B2B SKUs in <7 seconds.

Multi-Lingual Business Rationales

Generates personalized, non-technical value rationales explaining expected ROI in the customer's native tongue.

One-Click Bundle Hand-Off

Recommended bundle instantly populates Quote Builder with exact SKUs, contract tenure, and promotional pricing.

Smart KYC Pipeline & DPDP Compliance

Verified Against All 12 Hackathon KYC Specimen Files (K01–K12):

- K03 (Pvt Ltd) & K12 (NE Proprietorship) APPROVED (94% & 91% Trust Score)
- K01, K02, K04–K11 accurately flagged & REJECTED with specific compliance causes

Algorithmic Verification

Mod-36 GSTIN checksum validation, PAN entity code checking, Laplacian variance blur detection, Hough skew detection.

Registry Cross-Verification

Real-time automated reconciliation against mock GSTIN (3,534 records) & Udyam (2,723 records) registries.

Signatory Governance

Validates board resolutions, MOA signatory authority, partner consents, and power-of-attorney documentation.

DPDP Act 2023 Compliance

AES-256 encryption at rest, TLS 1.3 in transit, explicit consent capture, purpose limitation, and full audit trails.

Partner CRM, ML Propensity & Transparent Quote Builder

Partner CRM & ML Propensity Engine

Trained Random Forest Classifier (AUC 0.645)

Trained on smb_leads.csv (5,000 leads) utilizing 12 engineered features: business sector, employee size, inquiry channel, engagement velocity, and city circle.

3-Tier Propensity Scoring

Categorizes every incoming inquiry into High, Medium, or Low Propensity with estimated probability of activation within 30 days.

Automated Next-Best-Action (NBA) Prescriptions

High Propensity: Priority RM phone call within 2 hours with bundled discount.

Medium Propensity: Automated WhatsApp catalog & customer case study.

Low Propensity: Nurture email sequence with educational video guides.

Sales Channel Enablement

Provides field channel partners with real-time conversion insights, pending document alerts, and commission visibility.

Quote Builder & Real-Time Order Tracker

Full 25-Product Enterprise Catalog

Connectivity, Cloud SaaS, Cyber Security, IoT, and Collaboration SKUs clearly priced with no hidden charges.

Automated GST & Bundle Discount Engine

Instant calculation of 18% GST (CGST + SGST / IGST) and applied volume savings so the customer sees the exact out-of-pocket figure.

Instant Digital Contract Generation

Creates tamper-evident contract agreement with digital e-sign option, eliminating paper couriers and physical signature delays.

Zero-Chase Real-Time Order Tracking

Customer receives SMS/WhatsApp order link with 5 live milestones (Order Placed → Feasibility → KYC → Hardware Dispatch → Line Live) and SLA alerts.

Business Impact, Unit Economics & 90-Day Execution Roadmap

Measurable Business Impact & ROI

Metric	Baseline	Target	Delta
Customer Acquisition Cost	₹15,000	₹2,450	**-83.7%**
Lead-to-Activation Latency	17.6 Days	< 24 Hours	**-95.0%**
Funnel Conversion Rate	7.36%	28.0%	**+3.8x**
Status Inquiries per Deal	3.0 Calls	0 Calls (Self-Serve)	**-100%**
6-Month Post-Sale Churn	11.2%	< 4.0%	**-64.3%**
Activated SMBs / 5k Cohort	368 MSMEs	1,400 MSMEs	**+1,032 MSMEs**

Annualized Value Creation for Airtel:

- CAC Savings: (₹15,000 - ₹2,450) × 1,400 activations = ₹1.75 Cr per 5,000 leads.
- Recurring ARR Boost: +1,032 incremental SMBs @ ₹2,500/mo avg = ₹3.10 Cr ARR.
- Scaled to 100,000 Annual Leads: ₹35+ Cr CAC reduction + ₹62+ Cr added ARR.

90-Day Execution Roadmap & Rollout

1 Days 1–30: Pilot in 5 Metro Circles

Deploy in Mumbai, Delhi-NCR, Bengaluru, Hyderabad, and Chennai. Train 150 Airtel Partner RMs; onboard first 1,000 SMBs; refine OCR models.

2 Days 31–60: Tier-2 Circle Expansion

Scale across 25 Tier-2 cities (Pune, Ahmedabad, Jaipur, Lucknow, Kochi, etc.). Integrate with Airtel BSS billing engine and CRM API.

3 Days 61–90: Pan-India General Rollout

Activate across all 22 Airtel telecom circles. Launch customer self-onboarding web portal alongside partner assisted channel.

EVALUATION JURY ACTION: EXPLORE DELIVERABLES

Watch Demo Video Walkthrough (Google Drive)
https://drive.google.com/file/d/1TwcPl2c8KbH7sJIPwkrbOD9QJjv4vQ7D/view?usp=drive_link

View Source Code & Architecture (GitHub)
https://github.com/saurabhborikar13/Airtel-sync